WEB APPLICATION DEVELOPMENT

COURSE OUTLINE 2026-2027 | Tentative instructional sequence

Coalinga High School | Instructor: Manuel Chagoyan

Advanced capstone course. Students connect Terminal/Git workflows and web design foundations to visual JavaScript, Canvas, and AI-assisted creation, progressing from understanding and modifying code to directing natural-language development for a business/community application. Students publish and maintain an online career portfolio, prepare for the annual Career Skills Challenge, and complete individual and team projects.

Planning document: units, order, depth, tools, and sample assignments may change with student progress and evolving technology. Units may overlap; keyboarding, professional habits, and quality checks recur throughout the year.

Units & sample assignments	Anchor & CA CTE standards	Student learning objectives
01 Developer workspace & professional routines macOS/files review, VS Code, CodePen, project folders and accounts; typing practice and professional-skills certificates continue.	**Anchor**: 4.0, 6.0, 7.0 **Pathway**: ICT C4.3	Set up and maintain an organized development workspace; use accurate file paths and submissions; practice safe, responsible, and efficient work habits.
02 Terminal & real file operations pwd, ls, cd, mkdir, touch, cat, mv, cp, rm; Terminal Quest, Terminal 2047, and teacher-selected practice; real folder/file tasks.	**Anchor**: 4.0, 5.0 **Pathway**: ICT C4.3	Navigate and manipulate actual files from Terminal; interpret relative and absolute paths; verify targets before changing or deleting files; transfer game practice to real workflows.
03 Git, GitHub & documentation Local repository practice: init, status, add, commit, log; clone, pull, push; basic branches and merges; Markdown README; Git games as reinforcement.	**Anchor**: 4.0, 5.0, 9.0 **Pathway**: ICT C4.11, C5.4	Record meaningful checkpoints, inspect changes, and synchronize local and remote work; document project purpose and use; describe how version control supports ownership and recovery.
04 HTML/CSS, Photoshop & portfolio review Semantic HTML, Flexbox, responsive layout, color and typography; layers/masks and optimized assets; Photoshop and social contract review; refresh and publish personal portfolio (publishing permission form required); Career Skills Challenge preparation.	**Anchor**: 2.0, 4.0, 11.0 **Pathway**: ICT C3.3, C6.3, C6.5, C7.5; AME A8.4	Build and refine a responsive portfolio; integrate appropriate graphics; evaluate readability, keyboard access, and content quality; describe the structure of project files.

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Build and apply | Continued planned sequence

Units & sample assignments	Anchor & CA CTE standards	Student learning objectives
05 JavaScript foundations with visible results let/const, numbers, strings, booleans, operators, arrays, and basic objects; script setup, console, DOM text/color changes.	**Anchor**: 5.0 **Pathway**: ICT C4.4, C4.6, C4.7	Store and update values; predict expressions and screen changes; inspect console output; describe how data and program state affect behavior.
06 Decisions & repetition if/else, else if, comparisons and logical operators; switch where useful; for and while loops, stopping conditions; traffic lights and repeated patterns.	**Anchor**: 5.0 **Pathway**: ICT C4.6, C4.9	Trace branches and iterations; select a control structure for a task; change conditions and repetition safely; describe why a program produces its visible result.
07 Functions, events & DOM interaction Functions, parameters, return values; click/input/keyboard events; scoreboard, theme switcher, or interactive quiz; debugging challenges.	**Anchor**: 2.0, 5.0 **Pathway**: ICT C4.5, C4.9, C5.6	Organize behavior into reusable functions; connect user input to state and interface updates; predict, modify, test, and debug a small interactive program.
08 Canvas graphics & animation Coordinates, shapes, color, arrays of objects; requestAnimationFrame, motion, boundaries, keyboard input, simple collision; bouncing ball or small game.	**Anchor**: 5.0, 10.0 **Pathway**: ICT C4.7, C4.9, C6.4	Draw and animate objects; use conditions, loops, functions, and events to control motion; track animation state; test and revise a manageable visual project.

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Integrate and demonstrate | Continued planned sequence

Units & sample assignments	Anchor & CA CTE standards	Student learning objectives
09 AI-assisted coding & application planning Read and modify generated HTML/CSS/JS; prompt with requirements, compare outputs, debug iteratively; small app or game prototype.	**Anchor**: 4.0, 5.0, 8.0 **Pathway**: ICT C2.5, C4.11, C5.4, C5.5	Break a task into requirements; evaluate generated code; make and record an intentional change; test against expected behavior and document assistance.
10 Publishing, accessibility & search visibility GitHub-to-Netlify or Cloudflare workflow; publishing permission form on file; deployment checks, semantic structure, titles/meta descriptions, alt text, contrast and keyboard testing; Lighthouse/axe as appropriate.	**Anchor**: 4.0, 5.0, 8.0 **Pathway**: ICT C3.3, C7.3, C7.5, C7.6	Publish and update an application; inspect deployment problems; perform manual and tool-assisted quality checks; repair issues and report evidence and limitations.
11 Career documents & professional portfolio Refresh resume, cover letter, references, career plan, and interview responses; add code/design samples, certificates, and project descriptions.	**Anchor**: 2.0, 3.0, 11.0 **Pathway**: ICT C4.11, C7.5	Present accurate evidence of technical and professional skills; connect career goals to training options; describe project decisions and individual contributions.
12 AI business application - culminating project Individual or team project using natural-language development to address a local business/community need; requirements, prototype, feedback, revision, accessibility/SEO checks, code handoff and presentation.	**Anchor**: 2.0, 5.0, 9.0, 11.0 **Pathway**: ICT C2.3, C2.5, C3.3, C5.4, C5.5, C7.5	Define a useful problem and audience; guide AI with clear requirements; test realistic workflows; revise from feedback; deliver a working application and document its limitations and maintenance needs.

Standards key & alignment. Anchor numbers refer to broad California CTE Knowledge and Performance Anchor Standards. ICT C = Software and Systems Development; AME A = Design, Visual, and Media Arts. Listed standards identify relevant instructional connections, including partial coverage; they do not certify mastery of every component.

Sources: California CTE ICT standards and AME standards. Planning draws on prior outlines, course ideas, and instructor direction. Technology Pathway hub: resources and prior-year instructional records. Actual instruction is documented separately in Google Classroom.